Korea Technology and Information Promotion Agency for SMEs

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Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Advanced manufacturing of nacre-like architectures using micro/nano layering with ternary fillers for thermal-runaway protection in lithium-ion batteries.			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery (√), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Chemicals		Mechanical · Materials	
	Medium Category	Small Category	Medium Category	Small Category
	Polymeric Materials	Polymer Processing Technologies	Precision Manufacturing Machinery	Press Machines
Concept of Proposed Technology	The proposed technology focuses on the development of a nacre-inspired ternary filler–based composite film incorporating graphene [2][4], expandable graphite, and cellulose, specifically designed to mitigate thermal runaway in lithium-ion batteries. Through controlled mixing, dispersion, and shear-induced alignment during film processing, the composite film forms a bio-mimicking brick-and-mortar architecture, in which plate-like graphene and expandable graphite act as rigid building blocks while cellulose and the polymer matrix function as an interfacial binding phase. This nacre-inspired structure simultaneously enhances thermal insulation, mechanical robustness, and fire-retardant performance by promoting efficient stress redistribution, crack deflection, and heat management under thermal abuse. The ternary fillers synergistically contribute to endothermic heat absorption, laser-activated expandable graphite–driven char formation, and stabilization of a			

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	thermally insulating barrier under extreme conditions.[4] This scalable, formulation-driven nacre-inspired composite-film approach provides a robust and manufacturable solution for next-generation lithium-ion battery safety at both the module and system levels.[1]
Rationale for Joint R&D Project	The development of nacre-inspired ternary filler–based composite films [5][8] for lithium-ion battery thermal-runaway protection requires precise control over filler dispersion, interfacial interactions, and processing-induced structural organization, as well as scalable manufacturing capabilities that cannot be fully realized at the laboratory scale alone. While academic research establishes the fundamental mechanisms governing the synergistic behavior of graphene, expandable graphite, and cellulose, laser processing is essential for the spatially selective activation of expandable graphite, enabling controlled expansion and the formation of localized, nacre-like thermally insulating and fire-retardant barriers.[1] This targeted activation allows precise regulation of thermal and structural responses without compromising the bulk integrity of the composite film. In addition, roll-to-roll (R2R) film processing expertise is critical to translate these formulation-driven, bio-inspired material concepts into industrially viable, large-area formats. Korean roll-to-roll film manufacturing companies possess advanced capabilities in continuous coating, composite compounding, thickness uniformity control, and high-throughput production, and increasingly integrate inline or hybrid laser-processing technologies compatible with R2R manufacturing. These capabilities are crucial for implementing laser-triggered expandable graphite expansion and maintaining nacre-inspired filler organization within ternary composite films at industrial scale.[1][4][5] This collaboration enables the seamless integration of composite formulation design, nacre-inspired structural engineering, laser-induced expandable graphite activation, and industrial-scale R2R manufacturing, ensuring compatibility with existing lithium-ion battery module and pack assembly lines. Therefore, this joint R&D project is necessary to bridge the gap between laboratory-scale material innovation and large-area, cost-effective, and reliable battery safety films, thereby accelerating commercialization and strengthening global competitiveness.
Key Words	1. Bio-inspired design 2. Micro/nano-layered architecture 3. Ternary hybrid system 4. Nacre-like structure 5. Laser processing

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (2)	End of the Project	Level (6)
R&D Objectives	The primary objective of this joint R&D project is to develop scalable, bio-inspired nacre-like protective films based on ternary filler composite design [2][3] and processing-induced structural organization, enabling effective mitigation of thermal runaway in lithium-ion batteries. Specifically, this project aims to achieve the following objectives: **1st Stage: Basic Theories and Experiments** Establish a fundamental understanding of nacre-inspired brick and mortar architectures and their structure-property relationships for thermal protection. Investigate theoretical heat-transfer mechanisms and flame-retardancy principles, and explore interactions among plate-like graphene, expandable graphite, cellulose, and polymer matrices. **2nd Stage: Proof of Concept, Such as Practical Ideas or Patents** Design nacre-inspired composite film structures demonstrating synergistic thermal insulation, mechanical robustness, and fire-retardant functionality. Establish proof of concept for laser-triggered expandable graphite expansion and generate intellectual property through patent applications. **3rd Stage: Verification of Basic Performance at a Laboratory Scale** Fabricate small-scale nacre-inspired composite film samples and verify basic thermal insulation, mechanical properties, and fire-retardant behavior. Demonstrate laser-induced expandable graphite expansion and validate fundamental performance under simulated thermal abuse scenarios. **4th Stage: Evaluation of Key Performance of Materials, Components, and Systems at a Laboratory Scale** Develop and optimize ternary micro/nano filler systems while systematically evaluating effects of filler dispersion, alignment, and interfacial bonding on performance. Elucidate structure–processing–performance relationships through comprehensive characterization and laboratory-scale battery module testing. **5th Stage: Production and Performance Evaluation of Prototypes of Finalized Materials, Components, and Systems** Establish roll-to-roll manufacturing processes and produce prototype-scale films with finalized compositions. Validate prototype performance under battery-relevant conditions and demonstrate system-level applicability with existing manufacturing infrastructures. **6th Stage: Production and Performance Evaluation of Prototypes at a Pilot Scale** Scale up to pilot production with high-throughput roll-to-roll manufacturing and validate performance in full battery pack assemblies. Demonstrate seamless industrial integration, conduct field-testing with partners, and establish quality control protocols for commercialization of readiness.			
R&D Target Specifications	This R&D project aims to develop advanced thermal-runaway barrier films based on bio-inspired nacre-like composite architectures incorporating carbon-based structural frameworks.[3][5] The nacre-inspired structure is realized through controlled dispersion, alignment, and interfacial organization of graphene, expandable graphite, and cellulose within a polymer matrix, rather than conventional multilayer stacking. [8][9] The target performance specifications are defined as follows.			

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	1. Thermal resistance and high-temperature stability The developed films will exhibit exceptional thermal resistance and high-temperature stability, maintaining structural integrity without penetration or catastrophic failure under direct flame exposure or localized thermal shock conditions. The laser-activated expandable graphite will generate a stable, expanded char barrier, achieving a char yield of ≥ 60–70% after severe thermal abuse. The objective is to increase the existing heat resistance temperature from 600°C to 3500°C. The effective thermal resistance of the charred structure will correspond to ultra-high-temperature shielding behavior required for thermal-runaway protection. **2. Thermal-runaway mitigation performance** The films will provide a thermal-runaway delay time of ≥ 15 minutes at the cell and module levels, representing more than a twofold improvement compared to conventional barrier films. During thermal-runaway events, the maximum backside temperature rise will be limited to ≤ 200–250 °C, enabled by the synergistic effects of endothermic heat absorption, expandable graphite expansion, and nacre-inspired heat-flow disruption. **3. Film architecture and thickness control** The total film thickness will be controlled within the range of 100–300 μm to ensure compatibility with lithium-ion battery module integration. The nacre-inspired architecture will be governed by filler alignment length scales ranging from sub-micrometer to several tens of micrometers, with thickness uniformity across large-area films maintained within ±5% through roll-to-roll processing control. **4. Mechanical durability and structural integrity** The films will retain at least 80% of their original tensile strength after high-temperature exposure at ≥ 600 °C, owing to the crack-deflecting and stress-redistributing nature of the nacre-inspired brick-and-mortar structure. In addition, the films will demonstrate flexural durability exceeding 10,000 bending cycles at a bending radius of ≤ 5 mm without macroscopic cracking or mechanical failure. **5. Manufacturability and scalability (roll-to-roll processing)** The fabrication process will support continuous roll-to-roll processing at line speeds of ≥ 5–10 m min^{-1}, with scalable film widths of ≥ 300–500 mm. Defect density, including pinholes and local thickness nonuniformity, will be controlled to less than 1 defect per m^2, ensuring industrial-level yield and reliability. **6. Cost efficiency and system compatibility** The increase in material and processing costs will be limited to ≤ 20% compared to conventional barrier films, despite the significantly enhanced performance. The developed films will be fully compatible with existing lithium-ion battery module and pack assembly processes, requiring no additional curing steps or high-temperature post-treatments beyond inline laser activation of expandable graphite.
R&D Plan	This R&D project will be executed through a phased research strategy integrating bio-inspired nacre-like structural design [8][9], ternary filler composite optimization [4], laser-triggered expandable graphite activation, and roll-to-roll–compatible manufacturing. The overall approach is designed to establish clear structure–processing–performance relationships while enabling scalable and industrially viable production of advanced thermal-runaway protective films. In the initial phase, bio-inspired nacre-like composite architectures will be designed based on controlled dispersion, alignment, and interfacial organization of graphene, expandable graphite, and cellulose within

a polymer matrix.[8][9][10] Ternary filler systems will be formulated to achieve synergistic thermal insulation, flame retardancy, and mechanical robustness through brick-and-mortar–type structural organization.[4] Material dispersion, rheological behavior, shear-induced alignment, and interfacial compatibility will be systematically investigated through laboratory-scale composite compounding and coating experiments.

In the second phase, laser processing strategies will be developed specifically to enable localized activation of expandable graphite within the nacre-inspired composite films.[5][8][10] Controlled laser irradiation will be employed to induce selective expandable graphite expansion and in-situ formation of thermally insulating and fire-retardant char barriers.[1][6] Laser parameters, including power, scanning speed, and irradiation pattern, will be optimized to achieve reproducible expansion behavior while preserving the bulk mechanical integrity and large-area uniformity of the films.[1]

In the third phase, roll-to-roll–compatible composite coating and inline laser-assisted processing will be integrated to fabricate large-area films with uniform thickness and low defect density. Process parameters such as coating speed, shear conditions, drying or curing profiles, and inline laser activation conditions will be optimized to ensure continuous manufacturability and compatibility with industrial roll-to-roll production lines.

In the subsequent phase, the thermal, mechanical, and fire-retardant performance of the fabricated films will be quantitatively evaluated under conditions relevant to lithium-ion battery thermal runaway. Key performance metrics—including thermal stability, heat propagation suppression, expandable graphite expansion behavior, char formation efficiency, and mechanical durability—will be systematically correlated with composite structure, filler organization, and laser processing conditions to refine the overall design.

In the final phase, system-level validation will be conducted using lithium-ion battery module–relevant configurations to assess integration feasibility and performance reliability. In parallel, manufacturing scalability, process yield, and cost–performance trade-offs will be analyzed to define an optimized roll-to-roll and laser-assisted manufacturing window suitable for industrial deployment. Through this phased approach, the project will progress from laboratory-scale validation to pilot-scale demonstration, supporting rapid translation toward commercialization.

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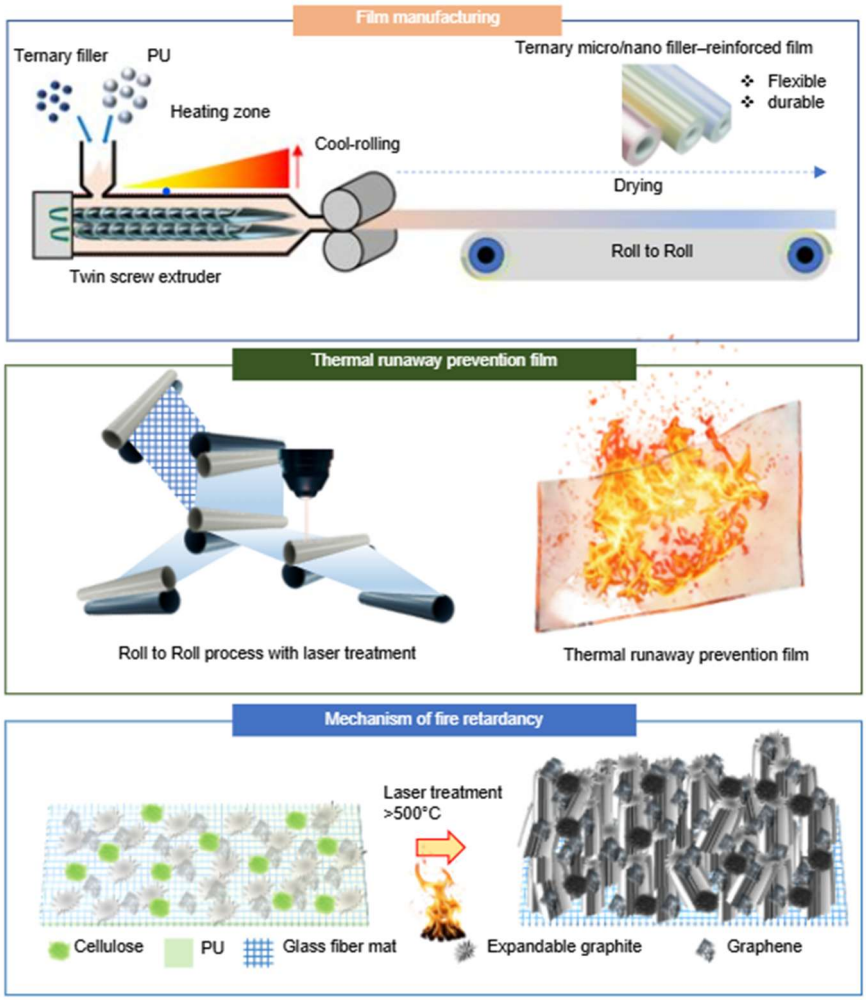

Figure 1. Schematic illustration of roll-to-roll fabrication, laser treatment, and fire-retardant mechanism of ternary filler-reinforced PU films.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The research institute at the University of Toronto will be responsible for the core scientific and technological development of the proposed project, with a focus on: **1. Bio-inspired nacre-like composite architectures** **2. Ternary filler systems for advanced thermal-runaway mitigation** From a technical contribution perspective, the research institute will lead the conceptual design and structural engineering of nacre-inspired composite films, in which plate-like graphene and expandable graphite are integrated with cellulose and polymer matrices to form brick-and-mortar–type architectures. These efforts will establish fundamental structure–function relationships governing stress redistribution, crack deflection, heat-flow disruption, and fire-retardant behavior, enabling synergistic thermal insulation, mechanical robustness, and thermal-runaway resistance. Within the research scope, the institute will develop and optimize ternary micro/nano filler systems [7], including filler selection, dispersion strategies, interfacial compatibility, and processing-induced alignment within polymer matrices. Systematic investigations will be conducted to elucidate structure–

	composition–property relationships governing endothermic heat absorption, laser-activated expandable graphite expansion, char formation, and suppression of heat and flame propagation. The research institute will also perform laboratory-scale fabrication and comprehensive characterization of nacre-inspired composite films to validate material concepts prior to industrial scale-up. This includes microstructural analysis, thermal and mechanical testing, and mechanistic evaluation under controlled thermal-abuse conditions relevant to lithium-ion battery safety. In terms of resources, the research institute will provide specialized research infrastructure, advanced characterization facilities, and experienced research personnel, including faculty members, postdoctoral researchers, and graduate students. These resources will support material design, experimental validation, and data-driven optimization of composite architectures and processing conditions. Furthermore, the research institute will deliver design guidelines, performance benchmarks, and technical documentation to support industrial partners during roll-to-roll manufacturing and laser-assisted expandable graphite activation development. Close collaboration with RISEN P&M and the Korean SME partner will ensure effective knowledge transfer and alignment between fundamental research outcomes and industrial implementation requirements.
Expected R&R of Korean SME Partner	RISEN P&M, as the Korean SME partner, will play a key role in the development, scale-up, and industrial validation of the proposed micro/nano-layered, bio-inspired nacre-like protective films. The company will leverage its expertise in roll-to-roll film manufacturing, polymer composite processing [2], and laser-assisted functionalization to translate research outcomes into industrially viable technologies. From a technical contribution standpoint, RISEN P&M will lead the process development and optimization of roll-to-roll–compatible coating and lamination techniques. This includes establishing stable process windows for multilayer stacking, thickness uniformity control, interlayer adhesion, and defect minimization under continuous production conditions. The company will also contribute to formulation refinement to ensure compatibility between material design and large-scale processing. RISEN P&M will be responsible for the integration and optimization of laser processing technologies within roll-to-roll manufacturing lines. This includes laser-based localized activation, functional patterning, and structural modification of the films to enhance thermal and fire-retardant performance. Laser parameters such as power density, scanning speed, and irradiation strategies will be systematically optimized to achieve reproducible, scalable processing.[1] In terms of testing and validation, RISEN P&M will conduct pilot-scale performance evaluation and manufacturability assessments, including thermal stability, flame resistance, mechanical durability, and process reliability tests. These activities will generate critical data to support iterative optimization and pre-commercial validation. RISEN P&M will commit essential industrial resources to the project, including roll-to-roll coating and lamination equipment, laser processing systems, quality control and in-line monitoring tools, and experienced technical personnel. These resources will support pilot-scale fabrication, process validation, and technology readiness advancement. Furthermore, RISEN P&M will actively collaborate with the research institute and other project partners, participating in joint technical meetings, data sharing, and integration studies with downstream battery module and system manufacturers. Through these collaborative activities, RISEN P&M will help accelerate scale-up, reduce commercialization risks, and support the transition of the developed technology toward industrial deployment.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	This joint R&D project is expected to deliver significant technological, industrial, and economic impacts through the development of advanced micro/nano-layered, bio-inspired nacre-like protective films for lithium-ion battery thermal-runaway mitigation. From a technological perspective, the project will establish a new class of multifunctional thermal-runaway barrier films that combine hierarchical micro/nano-layer architectures with optimized ternary filler systems.[4] Key achievements will include enhanced thermal stability, delayed thermal-runaway propagation, improved fire-retardant performance, and maintained mechanical robustness, all achieved through scalable roll-to-roll and laser-assisted manufacturing processes. The project aims to advance from the current baseline system to an improved target system with the following transformative improvements: Film Structure: Evolving from a 3-layer structure of polymer, expandable graphite (EG), and polymer to an advanced 3-layer structure of polymer and carbon polymer, enabling superior thermal management and structural integration. Applied Technologies: Advancing from perforation technology alone to an integrated multi-functional approach combining perforation technology, internal filling technology, and carbon structure technology—representing a technological complexity and capability for enhanced thermal protection. Cost Efficiency: Reducing production costs from high to relatively low through optimized manufacturing processes, targeting a significant cost reduction that will improve commercial viability and market accessibility. Heat Resistance Temperature: Dramatically increasing thermal resistance from 600°C to 3,500°C, representing an approximately 483% improvement (nearly six-fold enhancement) in high-temperature performance. This breakthrough will enable operation in extreme thermal environments previously unattainable with conventional thermal barrier coatings. Heat-Resistant Structure: Transitioning from conventional glass fiber (EG) materials to advanced carbon structure materials with superior thermal properties, thermal conductivity resistance, and structural stability under extreme conditions. Thermal Runaway Delay Time: Extending protection duration from 7 minutes to 15 minutes, representing a 114% improvement (more than doubling) in critical safety time. This enhanced delay provides substantially greater safety margins for emergency response, evacuation procedures, and fire suppression interventions. The project will also generate fundamental design rules and structure–processing–performance relationships that can be extended to other high-temperature and safety-critical applications. In terms of potential commercialization, the proposed technology addresses a rapidly expanding global market. The heat-resistant release film market is valued at $6.81 billion in 2025 and is expected to reach $16.81 billion by 2033 at a 11.95% CAGR, while high-performance heat-resistant polyimide films are projected to reach $3.04 billion by 2033, growing at a 9.12% CAGR. Heat-resistant packaging films are estimated to reach $6.7 billion by 2033. [11][12][13][14] This substantial growth is driven by surging demand from electronics, automotive, and industrial packaging sectors, with the Asia-Pacific region dominating due to rapid manufacturing expansion. The proposed technology is highly compatible with existing lithium-ion battery module and pack manufacturing infrastructures, enabling rapid adoption without major process modifications. The use of roll-to-roll manufacturing and laser processing supports large-area, high-throughput, and cost-efficient production, making the developed films attractive for commercialization in electric vehicles, energy storage systems, and next-generation battery platforms. The involvement of industrial partners ensures a clear pathway from pilot-scale demonstration to mass production and market entry,

	positioning the technology to capture significant market share in this high-growth sector. From an industrial and economic standpoint, this project will contribute to strengthening the global competitiveness of the Korean battery materials and film manufacturing industry by enabling the development of high-value safety components. The technology is expected to reduce battery safety risks, improve system reliability, and lower downstream costs associated with thermal management and fire protection. Furthermore, the project will foster industry–academia collaboration, accelerate technology transfer, and support workforce development in advanced manufacturing and battery safety technologies, ultimately contributing to sustainable industrial growth and economic impact.

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization